COMPUTER HARDWARE COMPONENTS AND UPGRADATION

Practical Assignment

Topic: Desktop PC Components, HDD vs SSD, CPU/RAM/GPU and Gaming PC Upgrades

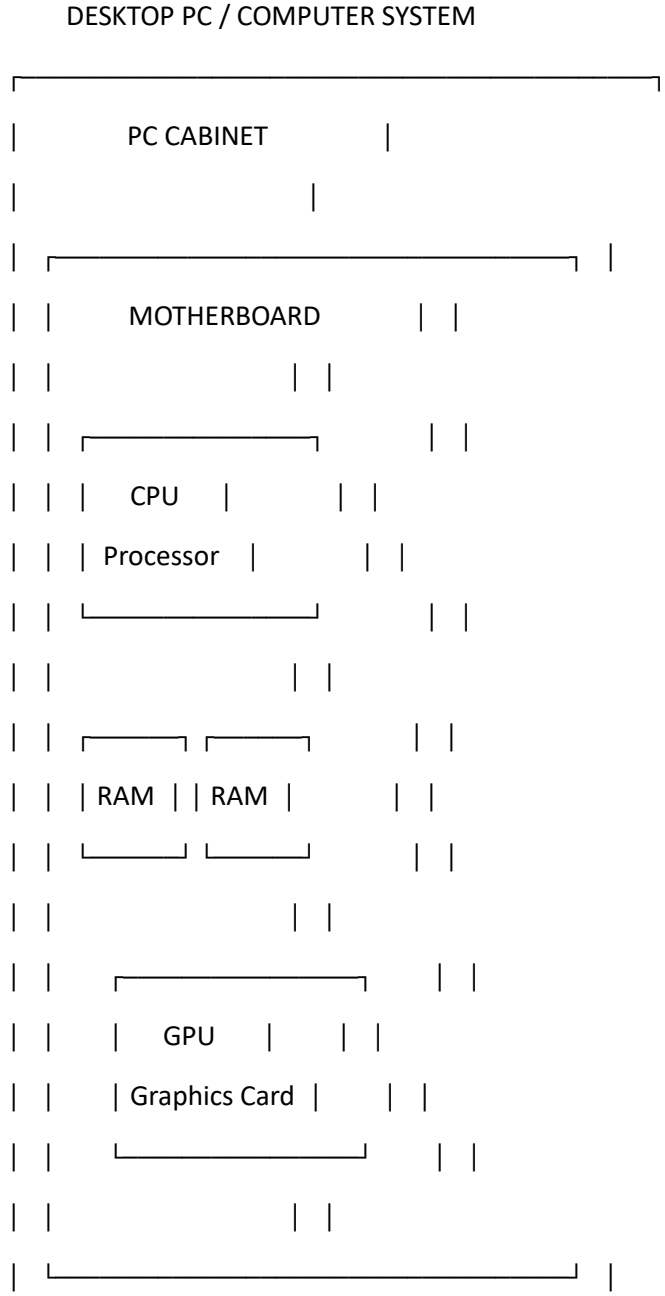
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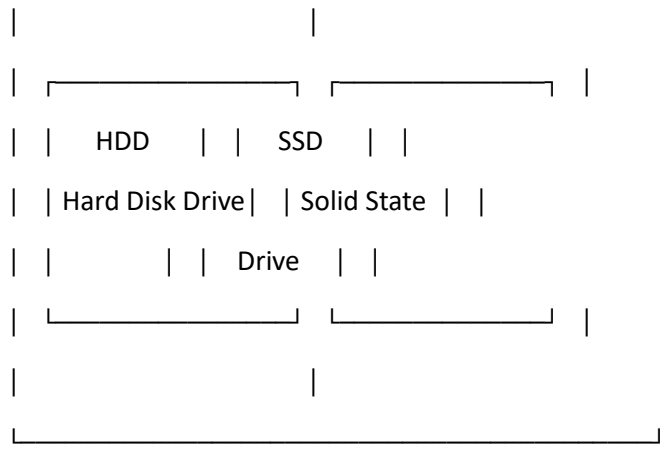
1. Labeled Diagram of a Desktop PC

Aim

To identify and understand the major hardware components of a desktop computer, including the motherboard, CPU, RAM, GPU, HDD and SSD.

Desktop PC Hardware Diagram





Description of Components

Motherboard

The motherboard is the main circuit board of a computer. It provides connections between the CPU, RAM, GPU, storage devices and other hardware components.

CPU

CPU stands for **Central Processing Unit**. It performs calculations and executes instructions given by computer programs and the operating system.

RAM

RAM stands for **Random Access Memory**. It temporarily stores data and instructions that are currently being used by the CPU and applications.

GPU

GPU stands for **Graphics Processing Unit**. It processes graphics and images and is especially important for gaming, video editing and other graphics-intensive tasks.

HDD

HDD stands for **Hard Disk Drive**. It stores files, applications and the operating system using magnetic disks and moving mechanical components.

SSD

SSD stands for **Solid State Drive**. It stores data using flash memory and has no moving mechanical parts, making it considerably faster and more resistant to physical shock than a traditional HDD.

Result

The major components of a desktop PC were identified and labeled, including the motherboard, CPU, RAM, GPU, HDD and SSD.

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2. Comparison Between HDD and SSD

Aim

To compare HDD and SSD storage based on speed, durability, technology, use cases and other important characteristics.

Comparison Table

Feature	**HDD**	**SSD**
Full Form	Hard Disk Drive	Solid State Drive
Storage Technology	Magnetic disks	Flash memory
Moving Parts	Yes	No
Speed	Generally slower	Generally faster
Boot Time	Longer	Shorter
Application Loading	Slower	Faster
File Transfer	Generally slower	Generally faster
Shock Resistance	Lower because of moving parts	Higher because there are no moving parts
Noise	Can produce mechanical noise	Silent
Power Consumption	Usually higher	Usually lower
Heat	Can generate more heat	Generally generates less heat
Weight	Generally heavier	Generally lighter
Price per GB	Usually cheaper	Usually more expensive
Capacity Options	Commonly available in large capacities	Available in many capacities
Common Use	Large storage, desktops, backups	OS drive, laptops, gaming PCs
Gaming	Good for storing games	Faster game loading
Laptop Use	Older/budget laptops	Modern laptops
Durability	More sensitive to physical shock	More shock-resistant
Performance	Suitable for general storage	Suitable for fast computing

Explanation

An HDD uses rotating magnetic disks and mechanical read/write components. Because of these moving parts, it is generally slower and more sensitive to physical shock.

An SSD uses flash memory and does not contain moving mechanical parts. This allows it to provide much faster access to data and makes it more resistant to physical movement and shock.

Typical Use Cases

HDD

HDDs are useful when a large amount of storage is required at a relatively low cost. They are commonly used for storing large collections of videos, photos, documents, backups and other files.

SSD

SSDs are commonly used as the main operating-system drive in modern laptops and desktops. They are also useful in gaming PCs because they can significantly reduce operating-system and game loading times.

Result

The comparison shows that HDDs are generally more economical for large-capacity storage, while SSDs provide faster performance and better resistance to physical shock.

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3. Role of CPU, RAM and GPU in a PC Game

Aim

To understand the functions of the CPU, RAM and GPU while running a game such as PUBG or Free Fire on a PC.

CPU – Central Processing Unit

The CPU handles the main calculations and instructions required by the game. It processes game logic, player actions, physics, AI, background tasks and communication between different hardware components.

For example, when a player moves, shoots or interacts with objects in the game, the CPU processes many of the instructions required for those actions.

RAM – Random Access Memory

RAM provides temporary working space for the game and the operating system. The game loads required data into RAM so that the CPU and other components can access it quickly.

Having sufficient RAM helps the computer keep the game, operating system and background applications running without excessive use of slower storage.

GPU – Graphics Processing Unit

The GPU is responsible for rendering the game's visual elements. It processes textures, lighting, shadows, characters, environments and other graphical effects before displaying them on the monitor.

A more capable GPU generally allows a PC to run graphics-intensive games at higher resolutions and graphical settings, subject to the game's requirements and the rest of the system.

Example During Gaming

When playing PUBG or Free Fire:

- **CPU** → Processes game logic and instructions.
- **RAM** → Temporarily holds game and system data.
- **GPU** → Creates and renders the game's graphics.
- **SSD/HDD** → Stores the game files and operating system.

Result

CPU, RAM and GPU work together to provide a smooth gaming experience. The CPU handles processing, RAM provides temporary working memory, and the GPU handles graphics rendering.

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4. Hardware Upgrades for Running the Latest FIFA Smoothly

Aim

To identify the hardware components that could be upgraded to improve gaming performance and run a modern FIFA/EA Sports FC title smoothly.

Introduction

Modern football games can require considerable graphics and processing performance. Before upgrading a computer, the existing hardware should be checked against the specific game's system requirements.

The most important upgrades generally involve the GPU, RAM, CPU and storage.

Components to Consider Upgrading

1. GPU – Graphics Card

The GPU is one of the most important components for gaming performance.

A stronger GPU can improve:

- Frame rates
- Resolution
- Texture quality
- Lighting and shadows
- Overall graphical quality

If the current graphics card is below the game's requirements, upgrading the GPU should be considered.

2. RAM

RAM provides temporary working memory for the game and operating system.

If the computer has insufficient RAM, the system may experience slowdowns when the game and other applications are running simultaneously.

Increasing RAM to an appropriate capacity can improve multitasking and help maintain smoother gameplay.

3. CPU

The CPU handles game logic, calculations and other processing tasks.

If the existing processor is significantly below the game's requirements, upgrading to a compatible and more powerful CPU can improve performance, especially in CPU-intensive situations.

Before upgrading, motherboard compatibility, CPU socket, BIOS support and cooling requirements should be checked.

4. SSD

An SSD does not necessarily provide the same direct frame-rate improvement as a GPU upgrade, but it can significantly improve storage performance.

Installing the operating system and game on an SSD can provide:

- Faster boot times
- Faster game startup
- Faster loading of game data
- Faster general system response

5. Power Supply Unit (PSU)

A more powerful GPU or CPU may require more power.

Therefore, the PSU should be checked before installing upgraded hardware. The PSU should provide sufficient wattage and the required power connectors for the selected components.

6. Cooling System

Gaming generates heat, particularly from the CPU and GPU.

If the existing cooling system is inadequate, upgrading the CPU cooler or improving cabinet airflow can help maintain appropriate operating temperatures.

7. Motherboard

The motherboard may not need to be upgraded in every case, but it can become necessary when the desired CPU, RAM or other components are incompatible with the existing motherboard.

Compatibility should be checked before purchasing new components.

Upgrade Priority Table

Component	Why Consider Upgrading?	Expected Benefit
GPU	Handles game graphics	Better graphics and frame rates
RAM	Provides working memory	Better multitasking and smoother operation

Component	Why Consider Upgrading?	Expected Benefit
CPU	Handles game processing	Better CPU-dependent performance
SSD	Faster storage	Faster loading and system response
PSU	Supplies power to components	Supports higher-power hardware
Cooling	Controls component temperature	Better thermal stability
Motherboard	Provides component compatibility	Allows compatible newer hardware

Important Precautions Before Upgrading

1. Check the exact system requirements of the game.
2. Check the current PC specifications.
3. Confirm motherboard compatibility.
4. Check the available RAM slots.
5. Check the power supply capacity.
6. Check whether the cabinet has enough space for the new GPU.
7. Check cooling requirements.
8. Back up important data before performing hardware changes.
9. Install the correct drivers after upgrading hardware.
10. Test the computer after installation.

Result

The GPU, RAM, CPU, SSD, PSU and cooling system were identified as important components to consider when upgrading a gaming PC. The exact upgrades should be selected according to the game's current requirements and the existing computer's hardware compatibility.

Overall Conclusion

This assignment provided an overview of important desktop computer hardware components and their roles in gaming.

The desktop PC diagram identified the motherboard, CPU, RAM, GPU, HDD and SSD. The HDD and SSD comparison demonstrated the major differences between traditional magnetic storage and modern solid-state storage.

The CPU, RAM and GPU were then explained in the context of running games such as PUBG and Free Fire. Finally, the hardware components that may need upgrading for a modern FIFA/EA Sports FC title were discussed, including the GPU, CPU, RAM, SSD, PSU and cooling system.

Understanding these components helps users make informed decisions when building, maintaining or upgrading a desktop gaming computer.

Learning Outcomes

After completing this assignment, the student should be able to:

1. Identify major components of a desktop PC.
2. Explain the purpose of a motherboard.
3. Understand the functions of CPU, RAM and GPU.
4. Explain the differ